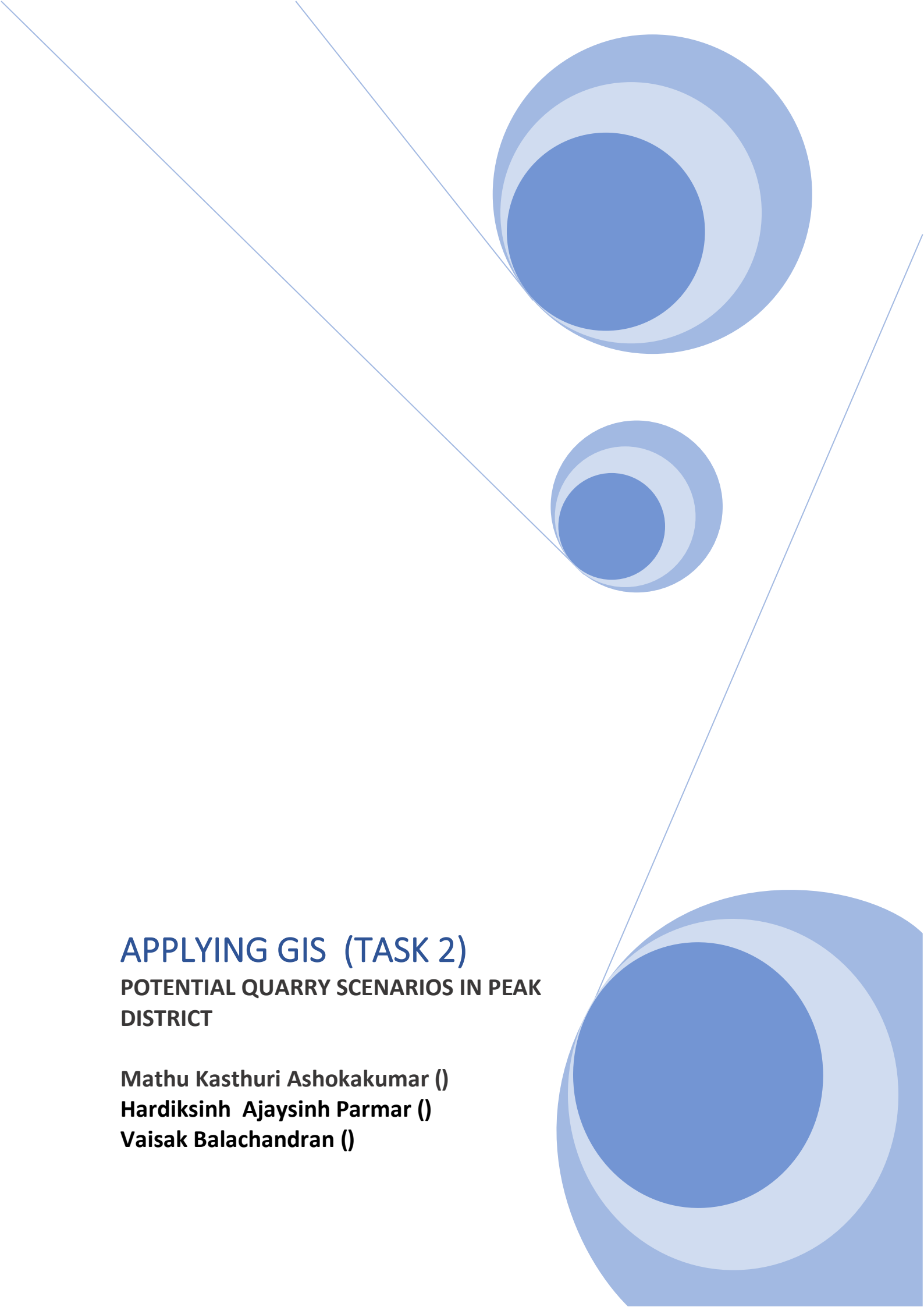




APPLYING GIS (TASK 2)

POTENTIAL QUARRY SCENARIOS IN PEAK DISTRICT

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INTRODUCTION

Mining studies encompass a wide range of research and analysis related to the extraction and processing of minerals and other resources from the earth. This can include studies related to the exploration, identification and evaluation of mineral deposits, the design and management of mining operations, and the assessment of the environmental, social and economic impacts of mining. Some examples of specific areas of focus in mining studies include:

Mineral exploration: Studies that focus on identifying and evaluating the potential of mineral deposits, including the use of geophysical, geochemical and remote sensing techniques.

Mining engineering: Studies that focus on the design and management of mining operations, including open-pit and underground mining, and the use of technology to improve efficiency and safety.

Environmental impact assessment: Studies that focus on evaluating the potential impacts of mining on the environment, including water resources, land use, and biodiversity.

Social impact assessment: Studies that focus on evaluating the economic and social impacts of mining on local communities, including job creation, economic development, and displacement of residents.

Health and safety: Studies that focus on improving the safety of mining operations and the health of workers, including the use of technology to monitor and manage risks.

Reclamation: Studies that focus on the restoration of mined lands to their pre-mining condition or to a useful post-mining land use.

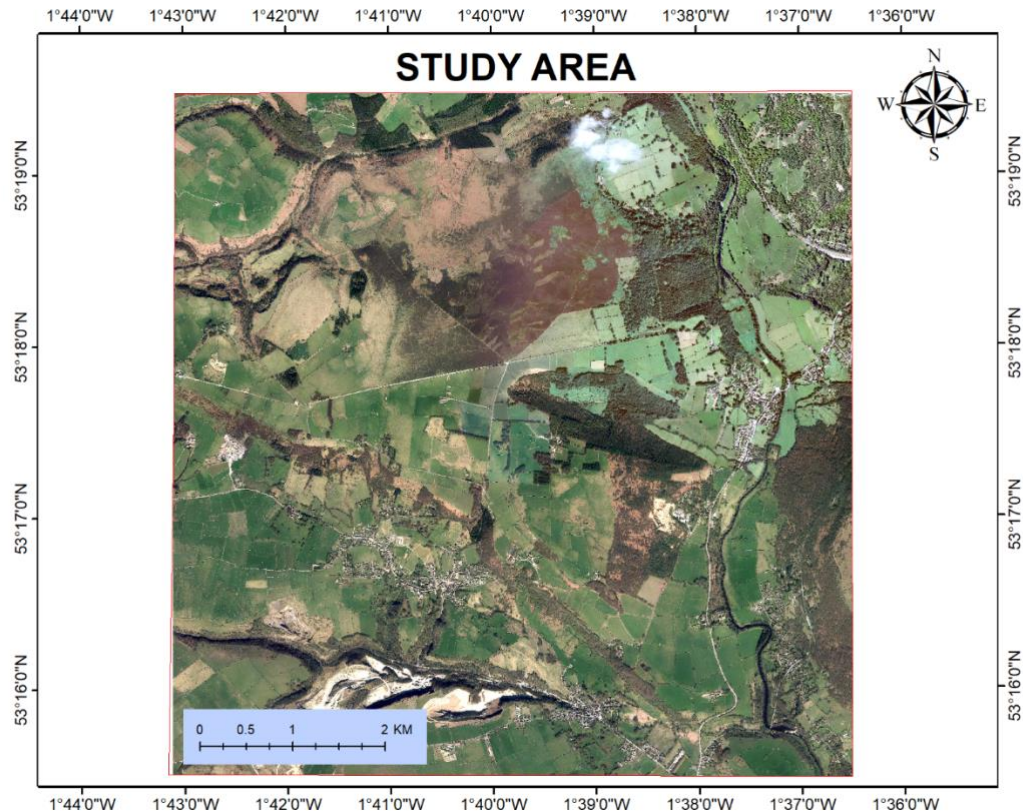
Mining Economics: Studies that focus on the economic aspects of mining, including the estimation of mining costs, the forecasting of mining revenues, and the analysis of mining investment.

Additionally, GIS can be used in mining to improve efficiency, reduce costs and improve safety, GIS can be used to manage and monitor resources, track equipment, and monitor environmental conditions.

STUDY AREA

Longstone Edge is a limestone escarpment in the Peak District of England. It is located in the Derbyshire Dales and is popular among hikers and rock climbers. The area is also known for its historical significance, including the Longstone, a large standing stone that stands at the edge of the escarpment. It is believed to have been used as a marker or boundary stone in ancient times.

Longstone Edge Quarry is a limestone quarry located near the village of Taddington in the Peak District of England. The quarry was in operation from the 19th century until the mid-20th century, and was used to extract high-quality limestone for use in construction and agriculture. The quarry has since been abandoned and the site has been reclaimed by nature. It is now a popular spot for hiking and rock climbing, with many well-worn footpaths leading to the quarry site. The area also provides a habitat for a variety of plants and animals, and is home to several species of birds of prey. The study area concentrates more on the north west part of the Longstone Edge.



METHODOLOGY

A 3D Analyst in ArcGIS is a software extension for ArcGIS that allows users to create and analyze 3D data and make 3D visualizations. It enables users to create 3D scenes, terrain models, and fly-through animations, as well as perform 3D spatial analysis and modeling. It is commonly used in fields such as GIS, surveying, engineering, architecture, and environmental studies.

Here are the basic steps to perform a slope map using 3D Analyst in ArcGIS:

- Open ArcMap and add the desired elevation raster to the map.
- Open the 3D Analyst toolbar by going to "View" > "Toolbars" > "3D Analyst."
- In the 3D Analyst toolbar, click on the "Surface Analysis" drop-down menu and select "Slope, Aspect, Contours Hill shade and View shed" one by one
- In the "Slope, Aspect, Contours Hill shade and View shed" dialog box, select the elevation raster as the input surface and choose the desired output measurement (e.g. degrees or percent).
- Click "OK" to create the slope map.
- You can then symbolize the map as desired, e.g. using a color ramp to show steeper and flatter areas.

Note: The above steps assume you have the 3D Analyst extension enabled. If not, you can enable it by going to the "Customize" menu and selecting "Extensions" then checking the 3D Analyst option and clicking "OK".

Also, note that you may need to prepare your data prior to the above process, depending on the format and resolution of your elevation data.

ArcScene is a 3D visualization software developed by Esri, a company that specializes in geographic information systems (GIS) software. It is a component of the ArcGIS Desktop software suite and allows users to create and view 3D scenes and analyze spatial data in a 3D environment. ArcScene can be used to create 3D visualizations of terrain, buildings, and other features, and can also be used to perform 3D analysis, such as viewshed analysis and terrain analysis. It also can be used to create 3D maps with 3D models and 3D symbols to represent real-world features.

The steps for creating a photomontage in ArcGIS using ArcScene include:

- Obtain a set of overlapping photographs or images of the scene you want to create. These images should be georeferenced, meaning they should have geographic coordinates assigned to them.
- Import the images into ArcScene by using the "Add Data" button.
- Use the "Align Photos" tool to align the images in 3D space. This tool uses the georeferenced coordinates of the images to match their positions and orientations in the 3D scene.
- Use the "Build Surface" tool to create a 3D surface from the aligned images. This tool uses the alignment information to create a surface that represents the terrain or other features in the scene.
- Adjust the lighting and other settings in the 3D scene, such as adding 3D models, changing the color of the surface, etc.
- Export the final output as a 3D scene, using the appropriate export tool.
- Note: Depending on the complexity of the scene, and the number of images, the process could be more complicated and the above steps might need to be repeated several times before getting a realistic 3D scene.

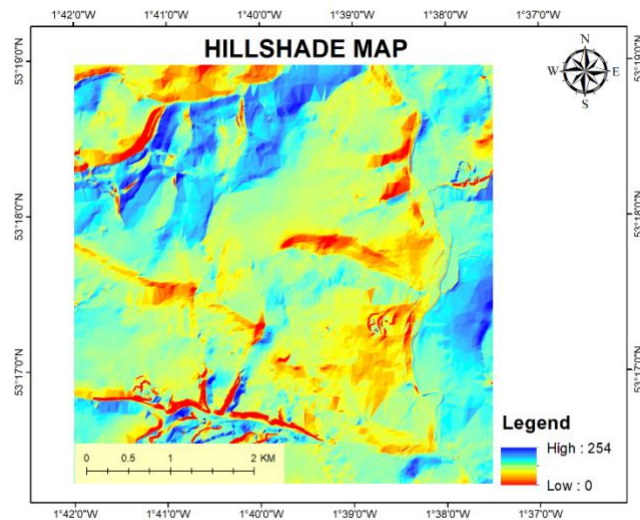
HILLSHADE ANALYSIS

A hillshade map is a type of topographic map that uses shading to represent the relief or elevation of the terrain. The shading is typically created by simulating the effect of sunlight falling on the landscape at a particular angle. This can help to give a sense of the three-dimensional shape of the terrain and make it easier to understand the topography of the area. Hill shade maps can be created using GIS software and digital terrain models (DTMs).

Step to perform hillshade :

- Open the DTM in a GIS software such as ArcGIS.
- Select the "Hillshade" or "Shaded Relief" tool, which can be found in the "Terrain Analysis" or "Raster Analysis" toolbox.
- Set the azimuth and altitude of the light source. Azimuth refers to the compass direction from which the light is coming, while altitude refers to the angle of the light source above the horizon.
- Set any other desired parameters such as the z-factor, which is used to adjust the vertical exaggeration of the relief, and the scale factor, which is used to adjust the overall brightness of the hillshade.
- Run the tool and the GIS software will generate a new raster layer that represents the hillshade.

- Add this layer to your map and adjust the symbology, transparency, or any other properties of the layer as necessary.
- Save the final map with the hillshade layer.



The benefits of using hill shade maps include:

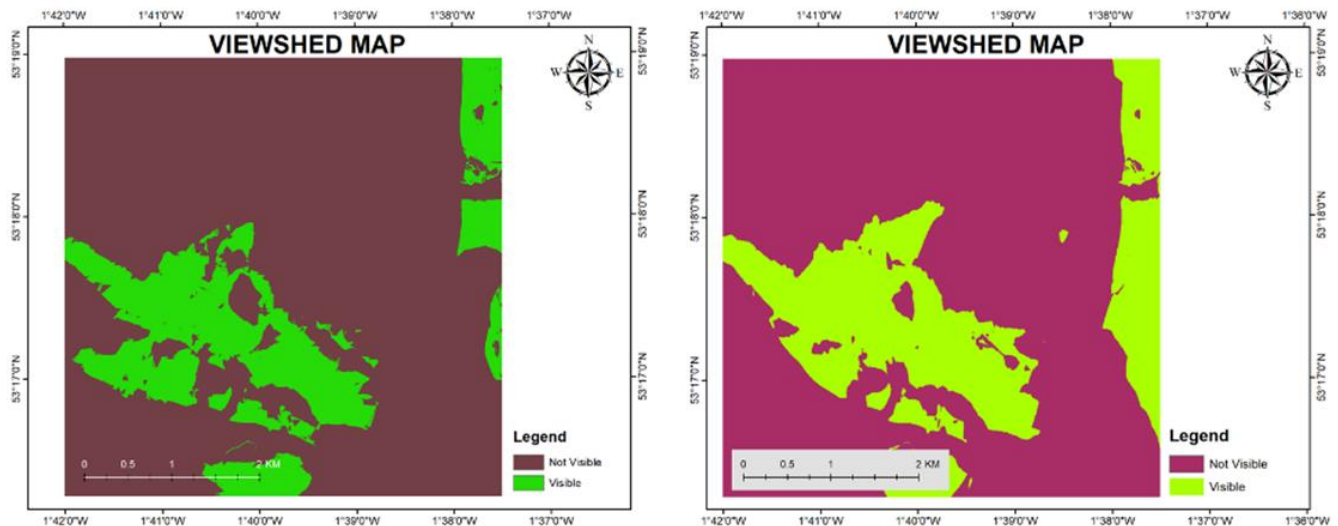
- Improved visualization: Hill shade maps provide a more realistic and intuitive representation of the topography of an area, making it easier to understand and interpret.
- Enhanced feature recognition: Hill shade maps can make it easier to identify and distinguish different topographical features, such as mountains, valleys, and ridges.
- Increased accuracy: By showing the relief of an area, hill shade maps can help to improve the accuracy of measurements and analyses that rely on topographical information.
- Better communication: Hill shade maps can be used to communicate topographical information more effectively, making them useful for a wide range of applications, including environmental studies, planning and development, and disaster management.

VIEWSHED ANALYSIS

Viewshed analysis in ArcGIS is a method of determining the visible area from a given location. It is used to identify areas that are visible from a given point, such as a lookout point or a tower.

Step to perform hillshade :

- Create a Digital Elevation Model (DEM): The first step is to create a Digital Elevation Model (DEM) of the area of interest. This can be done by using a variety of sources, such as LiDAR data, aerial imagery, or topographic maps.
- Create a Viewpoint Feature Class: The next step is to create a feature class that contains the viewpoint locations. This can be done by creating a point feature class and adding the coordinates of the viewpoint locations.
- Calculate Visibility: The final step is to calculate the visibility from the viewpoint locations. This can be done by using the Viewshed tool in ArcGIS. The Viewshed tool takes the DEM and the viewpoint locations as inputs and calculates the visible area from each viewpoint. The output is a raster layer that shows the visible area from each viewpoint.



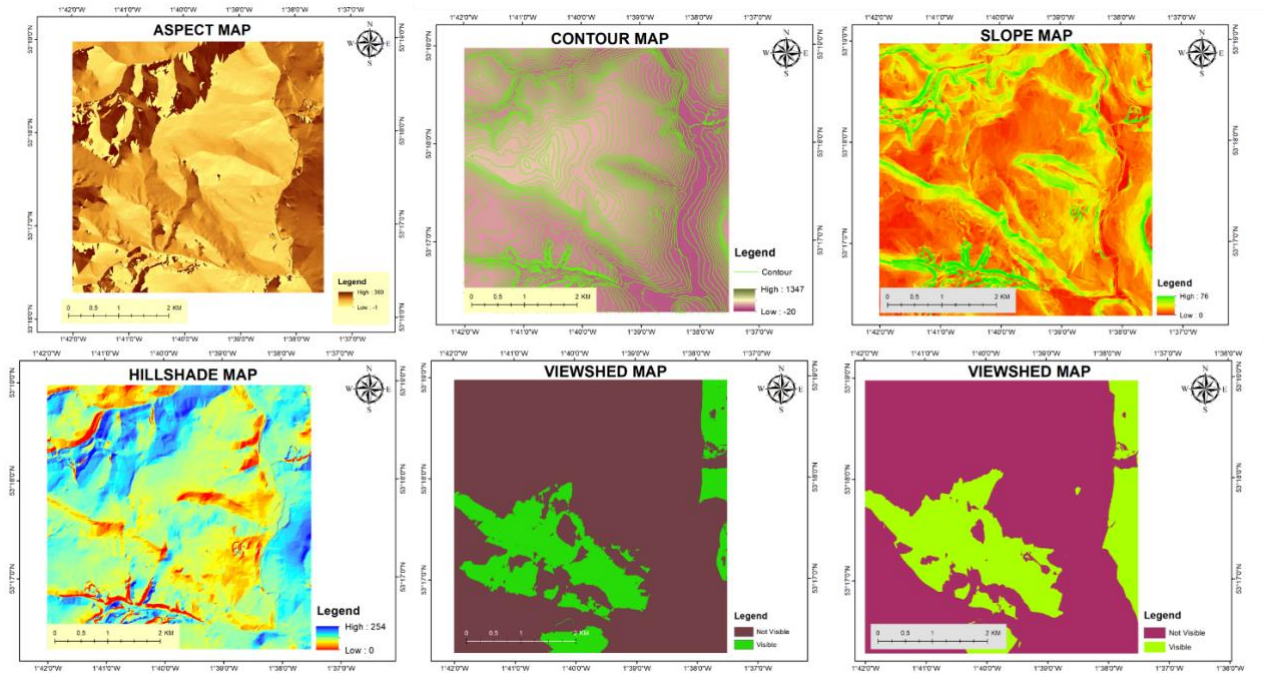
ANALYSIS AND RESULTS

Slope is a measure of the steepness of a surface or terrain. It is often represented as a percentage or as an angle, and is calculated by measuring the change in elevation over a certain distance. A slope of 0% indicates a flat surface, while a slope of 100% indicates a vertical surface.

In GIS, the slope can be calculated using DTM data. The most common method for calculating slope is by using the rise/run method, which calculates the slope as the ratio of the change in elevation (rise) to the distance (run) over which the change occurs.

It is worth noting that there are different ways to represent slope, such as degree, percentage, and gradient. It is important to use the correct representation for the specific application and to pay attention to the units of measurement.

An aspect map is a type of map that shows the direction of the slope of a terrain. It is created by calculating the aspect, which is the compass direction that a slope is facing. Aspect is typically represented as degrees from 0-360, with 0 being North, 90 being East, 180 being South, and 270 being West.



Contour lines are often labeled with the elevation at that particular point, this is called a contour interval, and the elevation difference between contour lines is constant. The distance between contour lines is called the contour interval, and this distance can be adjusted to suit the needs of the map and the terrain being represented.

The hillshade map is created by defining the position of the light source (usually the sun) and the direction of the surface normals (the upward direction of the surface). The algorithm then calculates the angle between the surface normal and the light source, and applies a shading value to each cell in the DTM based on this angle. Hillshade is a type of map that shows the relief of a terrain by simulating the shadows that would be cast by the sun at a specific time of day and date.

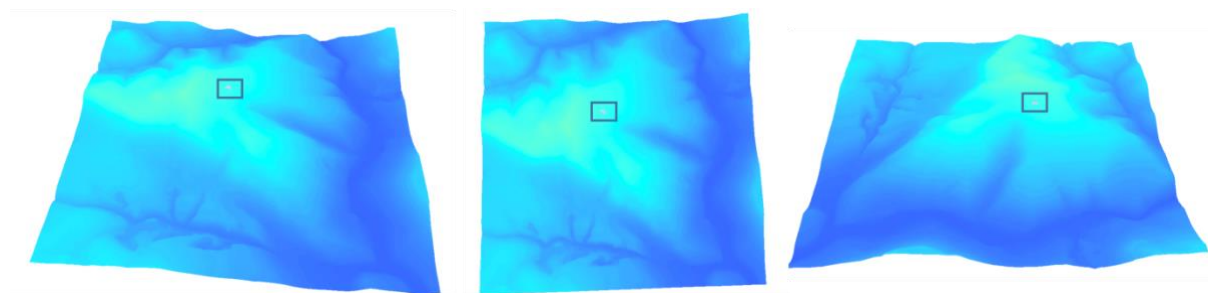
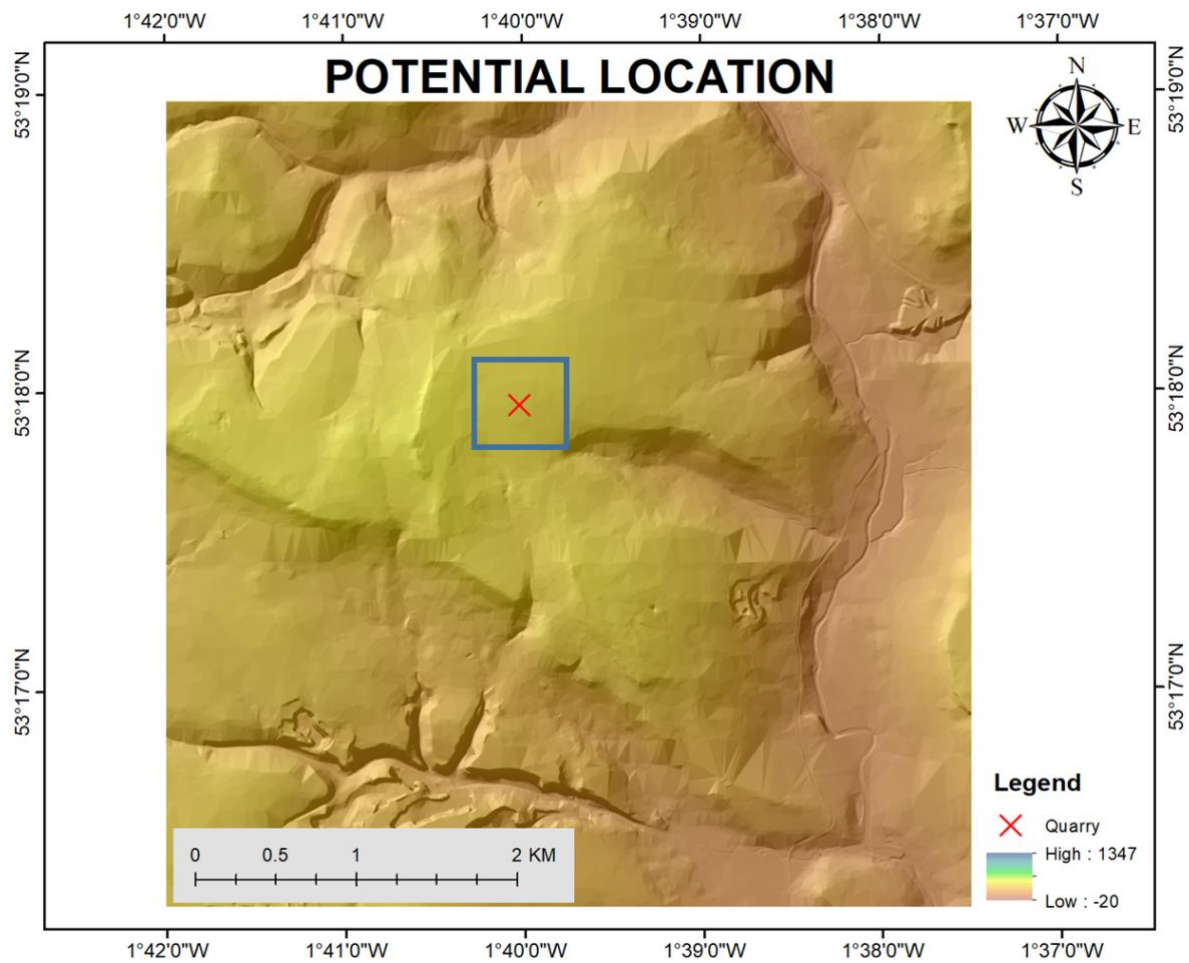
A viewshed is a map that shows the areas of a terrain that are visible from a specific location or set of locations. The result of a viewshed analysis is typically represented as a binary raster, where the areas that are visible are assigned a value of 1 and the areas that are not visible are assigned a value of 0. The result can also be represented as a continuous raster, where the value of each cell represents the visibility index, which is the degree of visibility from the observer's position.

The photo montage technique in ArcGIS is a method for creating a composite image of a 3D scene using multiple 2D photographs or images. This technique can be used to create realistic 3D visualizations of a scene or area, and can be used for a variety of purposes, such as planning, design, and analysis.

To create a photo montage in ArcGIS, you will need to have a set of overlapping photographs or images of the scene you want to create. These images can be taken from different angles and distances, and should be georeferenced, meaning they should have geographic coordinates assigned to them. Once you have your images, you can use the ArcScene software to import them and create a 3D scene by aligning the images and creating a 3D surface.

Additionally, you can use the tools in ArcScene to adjust the lighting, add 3D models, and perform other types of analysis, such as viewshed analysis and terrain analysis. The final output can be exported as a 3D scene. Overall, photo montage technique in ArcGIS is a powerful tool for creating

realistic 3D visualizations of a scene or area, and can be used for a variety of purposes, such as planning, design, and analysis.



Key factors of for proposing a new potential location are given below:

Slope: Steep terrain can make access and excavation difficult and costly, while flat terrain may be more suitable for large-scale surface mining operations.

Elevation: The elevation of a mining site can affect the climate, which in turn can affect the stability of the land and the feasibility of certain mining techniques.

Hillshade: The topography can dictate the accessibility of the mine, that is how easy or hard it is to reach the deposit, this can influence the cost of the mining operations.

Viewshed: The stability of the land and the feasibility of certain mining techniques.

CONCLUSION

Topography plays a significant role in mining as it affects the location, design, and cost of a mine. Steep terrain can make access and excavation difficult and costly, while flat terrain may be more suitable for large-scale surface mining operations. The presence of bodies of water or wetlands can also impact mining operations and may require additional permitting and mitigation measures. Additionally, the location of a mine in relation to population centers and environmentally sensitive areas can affect community and governmental approval of a mining project.

The GIS analysis of the Peak District National Park has provided a comprehensive overview of the potential quarrying scenarios. The hillshades, viewshed analysis, and 3D spatial analysis have provided a detailed understanding of the landscape and the potential impacts of quarrying. The results of this analysis have shown that quarrying in the Peak District National Park could have a significant impact on the landscape, environment, and local communities. Therefore, it is essential that any quarrying activities are carefully planned and managed to ensure that the impacts are minimized. This includes taking into account the potential visual, ecological, and hydrological impacts of quarrying, as well as the potential economic benefits. Furthermore, it is important to ensure that any quarrying activities are conducted in a sustainable manner, with the aim of preserving the natural beauty of the Peak District National Park for future generations.

PERSONAL REFLECTION

Mining and processing studies cover a wide range of topics. This includes study on mineral deposit exploration, identification, and appraisal, mining operation design and management, and the environmental, social, and economic impacts of mining. Mining studies focus on the following:

- Geophysical, geochemical, and remote sensing mineral exploration.
- The design, management, and use of technology to improve efficiency and safety in open-pit and underground mining.
- Environmental impact evaluations examine how mining may affect water, land, and biodiversity.
- Economic and social impacts of mining on local communities, including job creation, economic expansion, and resident relocation.
- Research to improve mine safety and worker health, including risk management technology.
- The study of returning mined land to its pre-mining state or to a beneficial use.
- Calculating mining expenses, projecting mining profits, and analysing mining investment.

The process of analyzing and interpreting geospatial data is crucial in understanding the potential impact of quarrying in an area of interest, such as the Peak District. In this report, we will discuss the steps taken to create aspect, contour, and hillshade maps, as well as an introduction to the potential quarrying scenario in the Peak District.

The first step in the process was to create an aspect map. This involved the use of a GIS software to create a map of the area of interest, and then using the software to calculate the aspect of each point on the map. The aspect of a point on a map represents the direction that a slope faces, with north-facing slopes having a different aspect than south-facing slopes. This information is crucial in understanding the potential impact of quarrying, as different aspects can affect the stability of the land and the feasibility of certain mining techniques.

The next step was to create a contour map. This involved the use of a GIS software to create a map of the area of interest and then using the software to calculate the elevation of each point on the map. The contour map helps to identify areas with different elevations, which can affect the climate and the stability of the land. It is also useful in identifying potential quarrying sites that are located at a suitable elevation for extraction.

The final step was to create a hillshade analysis. This involved the use of a GIS software to create a map of the area of interest and then using the software to calculate the slope of each point on the map. The hillshade analysis helps to identify areas of the Peak District that have a higher slope, which can make access and excavation difficult and costly, while flat terrain may be more suitable for large-scale surface mining operations.

In addition to creating the maps and analyses, an introduction was written for the report on potential quarrying scenario in the Peak District. This involved researching the area, understanding the context of the data, and writing a concise and informative introduction. It is crucial to understand the context of the data, as it allows for a better understanding of the potential impact of quarrying in the area of interest. The introduction provides a general overview of the Peak District and its geological characteristics, as well as the current state of quarrying in the area.

In conclusion, the process of creating aspect, contour, and hillshade maps, as well as an introduction to the potential quarrying scenario in the Peak District, is crucial in understanding the potential impact of quarrying in the area of interest. The use of GIS technology allows for the accurate analysis

and interpretation of data, which can then be used to identify potential quarrying sites and understand the feasibility of extraction. The information gathered through this process can be used to make informed decisions about quarrying in the Peak District and ensure that it is done in a responsible and sustainable manner.

GROUP CONTRACT

This contract is entered into by the following parties:

Party 1 : Mathu Kasthuri Ashokakumar

Party 2 : Hardiksinh Ajaysinh Parmar

Party 3 : Vaisak Balachandran

The parties agree to work together on a group assignment as follows:

Party 1 : Shall act as the group leader and shall be responsible for setting the agenda for the group meetings, organizing the meetings, and ensuring that all members of the group are informed of the meeting times and topics. She is also responsible for the analysis of results and creating of photomontage in ArcGIS and creating 3D models.

Party 2 : Data collection for the project, identifying major features in the study area and hill shade analysis in ArcGIS. Party 2 the major features identified, and the results of the hillshade analysis. And shall also be responsible for ensuring that the group is on track to meet its goals and objectives.

Party 3 : shall be responsible for extracting contours maps, viewshed analysis finalizing map layouts and also be responsible for the preparation of the final report, which shall include a summary of the data collected. A description of the methods used, and a discussion of the results. The report shall also include a detailed description of the contours maps, viewshed analysis, and map layouts.

Contract Clause :

1. All parties shall agree on a timeline for the completion of the Assignment.
2. All parties shall communicate regularly to ensure that the Assignment is progressing as planned.
3. All parties shall provide feedback and input to ensure that the Assignment is completed to the highest standards.
4. All parties shall be open to suggestions and ideas to ensure that the Assignment is completed to the best of their abilities.
5. All parties shall be willing to make adjustments and compromises to ensure that the Assignment is completed successfully.
6. All parties shall be accountable for their actions and decisions to ensure that the Assignment is completed in a timely and efficient manner.
7. All parties shall be respectful of each other's opinions and ideas to ensure that the Assignment is completed in a collaborative and productive manner.

This contract signed by:

Mathu Kasthuri Ashokakumar ()

Hardiksinh Ajaysinh Parmar ()

Vaisak Balachandran ()

Date: 19/01/2023

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